

# Lavneet Sidhu

 Lavneet Sidhu |  Lavneet.s.sidhu@gmail.com |  6476075932

## EDUCATION

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### Bachelor of Applied Science in Computer Engineering

September 2023 - June 2028

University of Toronto, Toronto, Ontario

- **Minors:** Artificial Intelligence, Robotics, Business
- **Relevant Courses:** Robot Modeling & Control (ECE470), Applied Fundamentals of Deep Learning (APS360), Introduction to Control Systems (ECE311), Operating Systems (ECE344), Algorithms and Data Structures (ECE345)

## PROFESSIONAL EXPERIENCE

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### Firmware Developer Intern

May 2026 - May 2027

Qualcomm

- Engineered virtual platforms for pre-silicon CPU and SoC modelling using **C/C++** and **SystemC**, optimizing hardware architecture evaluation for next-generation **AI/ML and robotics workloads**.
- Designed and integrated **transaction-level models (TLM)** to simulate processor architectures, bridging hardware simulation and real-time firmware execution for complex autonomous systems.
- Set up **CI/CD** pipelines and developed automated regression tests using **Python**, **Docker**, and **Git**, ensuring cycle-approximate accuracy against RTL reference designs for system determinism.

### Autonomous Drone Software Developer

September 2025 - May 2026

University of Toronto Aerospace Team (UTAT)

- Engineered autonomous aerial software stacks utilizing **ROS2**, **PX4 autopilot**, and **Gazebo simulation** to support robust flight navigation and control logic for competitive autonomous aerial missions.
- Collaborated with the computer vision sub-team to build **Edge AI** pipelines utilizing **YOLOv8** and **OpenCV** for camera-based **object detection**, alongside **LiDAR** integration on **NVIDIA Jetson Orin**.
- Contributed to the implementation of a comprehensive flight stack architecture for advanced mission planning and offboard autonomous control via the **uXRCE-DDS** interface operating on Linux-based Ubuntu.

### Machine Learning & LLM Researcher

May 2025 - Present

University of Toronto, Dalla Lana School of Public Health

- Optimized sentiment analysis and labeling pipelines by fine-tuning **LLMs** and traditional **NLP transformers**.
- Investigated and implemented **novel prompt engineering strategies** to significantly enhance LLM inference performance, with direct applications in large-scale sentiment analysis and programmatic tasks.
- Currently **authoring a comprehensive research paper** alongside Dr. Jude Kong at the Dalla Lana School of Public Health detailing recent artificial intelligence advancements for large-scale programmatic analysis.

## RELEVANT PROJECTS

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### Real-Time Autonomous Rover Tracking System

- Leading a team of 6 engineers building a computer vision-based rover tracking platform for autonomous navigation.
- Architected a sensor-fusion pipeline combining camera, LiDAR, and proximity sensors into a rover state vector.
- Integrated hardware sensing units with cloud-based storage and an interactive dashboard for post-processing analytics.

### Project Heimdall Counter UAS Detection Platform

- Engineered a sensor-agnostic drone-detection layer fusing radar, RF, acoustic, and EO/IR data into an air picture.
- Implemented multi-target tracking using Kalman filtering and nearest neighbour assignment across multiple sensors.
- Integrated ML classification models distinguishing threat drones from birds, auto-generating forensic geofence reports.

## TECHNICAL SKILLS

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- **AI/ML & Cloud:** PyTorch, TensorFlow, AWS, LLM Prompt Engineering, NLP, OpenCV, YOLOv8, ROS2
- **Programming Languages:** Python, C, C++, Verilog, SQL, R, Java, JavaScript, Node.js, React, MATLAB
- **Developer Tools:** Docker, Git, Eclipse, VS Code, Jupyter, ModelSim, MATLAB Simulink, Linux/Ubuntu
- **Soft Skills:** Leadership, Technical Communication, Public Speaking, Agile Methodologies, French

## HONORS & AWARDS

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### 1st Place Canadian Engineering Competition (Programming)

2025, 2026

- Swept 1st at the university (UTEK), provincial (OEC), and national (CEC) levels, topping 55 Canadian universities for U of T's first programming win in two decades.
- Engineered winning scalable cloud solutions under strict time constraints, including an AWS-hosted interplanetary navigation system and an AR defect-inspection platform utilizing Unity and DynamoDB.